

2023-2024 Fall
MAT123 Midterm
01/11/2023

1. Evaluate the following limits without using L'Hôpital's rule.

a. $\lim_{t \rightarrow 0} \frac{\tan^{-1} t}{\sin^{-1} t}$

b. $\lim_{x \rightarrow 0^+} x e^{\sin(1/x)}$

2. A spherical balloon is being filled with air in such a way that its radius is increasing at the constant rate of 2 cm/s. At what rate is the volume of the balloon increasing at the instant when its surface has area 4π cm²?

3. a. State MVT.

- b. Using MVT, show that for all $x > 0$,

$$1 + x < e^x < 1 + x e^x$$

4. Find the tangent line to the curve defined implicitly by the equation

$$y^2 x^x + xy = 2$$

at the point $(1, 1)$. Note that $y = f(x)$.

5. Find the number A so that $\lim_{x \rightarrow \infty} \left(\frac{x + A}{x - 2A} \right)^x = 5$.

6. Sketch the graph of $f(x) = (\ln x)^2$.

1. a. Rewrite the limit.

$$\begin{aligned}\lim_{t \rightarrow 0} \frac{\tan^{-1} t}{\sin^{-1} t} &= \lim_{t \rightarrow 0} \left(\frac{\tan^{-1} t}{\sin^{-1} t} \cdot \frac{t}{t} \right) = \lim_{t \rightarrow 0} \left(\frac{\tan^{-1} t}{t} \cdot \frac{1}{\frac{\sin^{-1} t}{t}} \right) \\ &= \lim_{t \rightarrow 0} \frac{\tan^{-1} t}{t} \cdot \frac{1}{\lim_{t \rightarrow 0} \frac{\sin^{-1} t}{t}} = 1 \cdot 1 = \boxed{1}\end{aligned}$$

The limits above are well-known limits. If we're supposed to write these limits in the form of $\sin x$ and x , follow these steps.

$$\begin{aligned}\lim_{t \rightarrow 0} \frac{\tan^{-1} t}{t} &\stackrel{u=\tan^{-1} t}{=} \lim_{u \rightarrow 0} \frac{u}{\tan u} = \lim_{u \rightarrow 0} \frac{\cos u}{\frac{\sin u}{u}} = \frac{\lim_{u \rightarrow 0} \cos u}{\lim_{u \rightarrow 0} \frac{\sin u}{u}} = \frac{1}{1} = 1 \\ \lim_{t \rightarrow 0} \frac{\sin^{-1} t}{t} &\stackrel{v=\sin^{-1} t}{=} \lim_{v \rightarrow 0} \frac{v}{\sin v} = \frac{1}{\lim_{v \rightarrow 0} \frac{\sin v}{v}} = \frac{1}{1} = 1\end{aligned}$$

- b. Start with the inequality below.

$$\begin{aligned}-1 &\leq \sin(1/x) \leq 1 \\ e^{-1} &\leq e^{\sin(1/x)} \leq e^1 \\ xe^{-1} &\leq xe^{\sin(1/x)} \leq xe \\ \lim_{x \rightarrow 0^+} xe^{-1} &\leq \lim_{x \rightarrow 0^+} xe^{\sin(1/x)} \leq \lim_{x \rightarrow 0^+} xe \\ 0 &\leq \lim_{x \rightarrow 0^+} xe^{\sin(1/x)} \leq 0\end{aligned}$$

By the squeeze theorem, the limit is $\boxed{0}$.

2. Let $V(t)$, $S(t)$, $r(t)$ represent the volume, surface area and radius, respectively. $r'(t) = 2$ for all t . The rate of change of volume at $t = t_0$ is

$$V'(t_0) = 4\pi r^2(t_0)r'(t_0) \quad \left[V(t) = \frac{4}{3}\pi r^3(t) \right]$$

We also have $S(t_0) = 4\pi$. Using the surface area formula $S(t) = 4\pi r^2(t)$, we find that at $t = t_0$, the radius of the balloon is 1. Therefore, the rate of change of volume can now be evaluated.

$$V'(t_0) = 4\pi \cdot 1^2 \cdot 2 = \boxed{8\pi \text{ cm}^3/\text{s}}$$

3. a. The MVT states that if a function f is continuous on $[a, b]$ and differentiable on (a, b) , there is at least one point P on the interval (a, b) such that the slope of the line that passes through the endpoints is equal to the slope of the line that is tangent to P
- b. Let $f(x) = e^x$. f is continuous on $[0, x]$ and differentiable on $(0, x)$. There exists at least one point c on $(0, x)$ such that

$$f'(c) = e^c = \frac{e^x - 1}{x} = \frac{f(x) - f(0)}{x - 0}$$

From the inequality $0 < c < x$,

$$\begin{aligned} e^0 &< e^c < e^x \\ 1 &< \frac{e^x - 1}{x} < e^x \\ x &< e^x - 1 < xe^x \\ 1 + x &< e^x < 1 + xe^x \end{aligned}$$

4. Let us find the derivative of x^x with respect to x .

$$\begin{aligned} y &= x^x \\ \ln(y) &= \ln(x^x) = x \ln x \\ \frac{1}{y} \cdot y' &= 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1 \\ y' &= x^x (\ln x + 1) \end{aligned}$$

We can now differentiate both sides of the equation given in the question.

$$\begin{aligned} \frac{d}{dx} (y^2 x^x + xy) &= \frac{d}{dx} 2 \\ 2y \cdot y' \cdot x^x + y^2 \cdot x^x (\ln x + 1) + 1 \cdot y + x \cdot y' &= 0 \\ y' \cdot (2y \cdot x^x + x) &= -y - y^2 \cdot x^x (\ln x + 1) \\ y' &= -\frac{y + y^2 \cdot x^x (\ln x + 1)}{2y \cdot x^x + x} \end{aligned}$$

Evaluating y' at $(1, 1)$ gives $-\frac{2}{3}$. Use the straight line formula $y - y_0 = m(x - x_0)$.

$$y - 1 = -\frac{2}{3}(x - 1)$$

5. Take the logarithm of both sides of the equation and apply L'Hôpital's rule.

$$\begin{aligned}
 \ln(5) &= \ln \left[\lim_{x \rightarrow \infty} \left(\frac{1 + \frac{A}{x}}{1 - \frac{2A}{x}} \right)^x \right] = \lim_{x \rightarrow \infty} \ln \left[\left(\frac{1 + \frac{A}{x}}{1 - \frac{2A}{x}} \right)^x \right] = \lim_{x \rightarrow \infty} \left[x \ln \left(\frac{1 + \frac{A}{x}}{1 - \frac{2A}{x}} \right) \right] \\
 &= \lim_{x \rightarrow \infty} \left\{ x \left[\ln \left(1 + \frac{A}{x} \right) - \ln \left(1 - \frac{2A}{x} \right) \right] \right\} \quad [\infty \cdot 0] \\
 &= \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{A}{x} \right) - \ln \left(1 - \frac{2A}{x} \right)}{\frac{1}{x}} \quad \left[\frac{0}{0} \right] \\
 &\stackrel{\text{L'H.}}{=} \lim_{x \rightarrow \infty} \left[\frac{\left(\frac{1}{1 + \frac{A}{x}} \right) \cdot \left(-\frac{A}{x^2} \right) - \left(\frac{1}{1 - \frac{2A}{x}} \right) \cdot \left(\frac{2A}{x^2} \right)}{-\frac{1}{x^2}} \right] \\
 &= \lim_{x \rightarrow \infty} \left[\frac{\left(-\frac{A}{x^2} \right) \cdot \left(\frac{1}{1 + \frac{A}{x}} + \frac{2}{1 - \frac{2A}{x}} \right)}{-\frac{1}{x^2}} \right] \\
 &= A \lim_{x \rightarrow \infty} \frac{1}{\frac{A}{x} + 1} + A \lim_{x \rightarrow \infty} \frac{2}{1 - \frac{2A}{x}} = A \cdot 1 + A \cdot 2 = 3A
 \end{aligned}$$

If $\ln(5) = 3A$, then $\boxed{A = \frac{\ln 5}{3}}$.

6. First off, find the domain. The expression is defined *only* for $x > 0$. The only vertical asymptote occurs at $x = 0$.

$$\mathcal{D} = \mathbb{R}^+$$

The limits as $x \rightarrow \infty$ and as $x \rightarrow 0^+$ are:

$$\lim_{x \rightarrow \infty} (\ln x)^2 = \infty, \quad \lim_{x \rightarrow 0^+} (\ln x)^2 = \infty$$

Take the first derivative to find the critical points.

$$y' = 2 \ln x \cdot \frac{1}{x}$$

The *only* critical point is $x = 1$.

Take the second derivative by applying the quotient rule.

$$y'' = 2 \cdot \frac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \frac{2 - 2 \ln x}{x^2}$$

The *only* inflection point is $x = e$.

Consider some values of the function. Eventually, set up a table and see what the graph looks like in certain intervals.

$$f(1) = 0, f(e) = 1$$

x	$(0, 1)$	$(1, e)$	(e, ∞)
y	$(0, \infty)$	$(0, 1)$	$(1, \infty)$
y' sign	-	+	+
y'' sign	+	+	-

